

1. Copy Windows 11 iso image to the Downloads folder
2. Install all the packages needed for creating virtual machines:

```
sudo apt install qemu-kvm libvirt-daemon bridge-utils virt-manager
```

3. Add ourselves to the Libvirt and KVM groups:

```
sudo usermod -aG libvirt $(whoami)
sudo usermod -aG kvm $(whoami)
```

4. Sign out and back in again, so group membership is recognized
5. Start Libvirt:

```
sudo systemctl start libvirtd
```

6. Enable it:

```
sudo systemctl enable
```

7. Install XfreeRDP

```
sudo apt install freerdp2
```

8. And then MenuLibre:

```
sudo apt install menulibre
```

9. Install Windows 11 Pro:

- Start VMM from the App Center
- Click on the leftmost icon at the top (Create a new VM)
- Select “Local Install Media” in the popup window
- Press the “Forward” button
- Press the “Browse” button
- Press the “Browse Local” button
- Navigate to and select the Windows 11 iso file in the Downloads folder
- Press the green “Open” button at the top of the Window
- Press the “Forward” button

- Use the defaults of 8192MB memory and 4 CPUs
- Press the forward button
- Use the default of 128GB disk size
- Press the “Forward” button
- Check the “customize configuration before installation” checkbox
- Press the “Finish” button
- Select “Boot Options”
- Check the “start virtual machine on host boot up” checkbox
- Click “Begin Installation” in the top left corner of the screen

<< A normal installation of Windows 11 will begin. When given the choice, you need to select “Windows 11 Pro”. The “Home” and “Educational” versions do not work for this purpose. These instructions pick back up when the Windows installation is complete. >>

10. Now, I hate Microsoft breathing down my neck. To sign into Windows, you need a Microsoft account. But, once you’ve signed in once, you can create a local account and cut the ties to Microsoft.
11. In Windows, create a new local account with no user credentials. Select the new account and change its type to Administrator
12. Sign out of Windows
13. Sign-in as the new user
14. Delete the original network account.
15. Windows runs better with fewer resources if we install special drivers for the virtual machine environment. These open source drivers are signed and distributed by Red Hat. Install “virtio-win-gt-x64.msi” and “virtio-win-guest-tools.exe” from the “virtio-win-<version number>” disk image

16. Determine the IP address of the Windows virtual machine. Right click on the network icon in the lower right of the home screen. Select "Network and Internet Settings". Click on Ethernet and scroll down to "IPv4 address.
17. We have to enable RDP on this copy of Windows. Use the Search box on the bottom middle of the home screen and search for exactly "Remote Desktop Setting". The correct entry will have an icon of a greater-than and less-than above each other. Choose this entry. Slide the switch on the upper right of the screen to "ON". Confirm you want to "Enable Remote Desktop".
18. Install the open source Windows application called RemoteApp Tool. After installation, the application will have an icon on the desktop. Launch the application. Click on the "Tools" menu across the top of the application window. Select "Host options...". Check the "Disable AppAllowListThis" checkbox. Check the "Allow Unlisted Remote Programs checkbox. Press the "Save" button and close the application.
19. Install the RDPWrapper application so we can run more than one Windows application at a time.
20. Download the "rdpwrap.ini" file from GitHub and copy it to "[C:\Program Files\RDP Wrapper](#)". Confirm you want to replace the .ini file already there. This new .ini file is configured specifically for Windows 11.
21. Power off Windows.
22. Let's take our new Windows application for a spin. It shouldn't require a reboot, but for some reason it does. So we reboot
23. This command line will launch Windows Notepad in Linux:

`xfreerdp /u:<name of the local account user in Windows> /p:<the password you gave that account> /v:<the IPv4 address of the Windows virtual machine> +dynamic-resolution +auto-reconnect`

+home-drive +clipboard -wallpaper /app:c:\\windows\\[notepad.exe](#)

24. It's a bit complicated which is why we want MenuLibre. We can configure an icon for each Windows application and bury the complex command line in the menu entry.